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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (Honours)/ MGU-BBA (Honours)/ MGU-BCA (Honours) REGULAR EXAMINATION October 2026

Fifth SEMESTER

Skill Enhancement Course (SEC) - MG5SECFTQ300 - FOOD PRODUCT DEVELOPMENT

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate€, Create(C), Skill(S), Interest(I) And Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A – Short Answer Questions (2 marks each)

1. State any two ways of identifying market opportunities.

Conduct market research, surveys and consumer interviews to identify unmet needs.
Analyse market trends, competitors and gaps in existing products.

2. Mention any two characteristics of a successful new food product.

It meets consumer needs and has acceptable taste, quality and convenience.
It is safe, affordable, commercially viable and differentiated from competing products.

3. Explain the importance of licensing in international trade.

Licensing permits a company to use another firm's brand, technology, formula or process under agreed terms.
It supports entry into foreign markets and defines usage rights, quality requirements and royalty payments.

4. Explain the importance of packaging in maintaining product stability.

Packaging protects food from moisture, oxygen, light, contamination and physical damage.
Suitable packaging helps preserve quality and extend shelf life.

5. Explain the importance of comparing market samples in food quality evaluation.

Comparison assesses colour, flavour, texture, appearance and acceptability against competing products.
It identifies quality gaps and opportunities for improvement.

6. Why should food companies consider market competition while developing new products?

Competitor analysis reveals existing product features, prices, strengths and market gaps.

It helps companies differentiate the product and plan suitable positioning and pricing.

7. State the role of market demand in food product development.

Market demand indicates whether consumers are likely to buy the proposed product.

It guides product features, production volume, pricing and launch decisions.

Part B – Short Essay Questions (4 marks each)

8. Describe how changing consumer preferences create opportunities for new food product development.

Changing preferences—such as demand for convenience, health, plant-based foods or sustainable packaging—reveal new consumer needs.

Companies can respond with suitable formulations, flavours, portion sizes and packaging.

Consumer research and product trials help confirm whether the product meets these needs.

9. Explain the role of market testing in reducing product launch risk.

Market testing introduces a product or concept to a limited target market before a full launch.

Feedback on taste, price, packaging and purchase intention reveals problems that can be corrected early.

Results help estimate demand and improve launch, distribution and promotional decisions.

10. Explain the importance of shelf-life evaluation of food products.

Shelf-life evaluation determines how long a product remains safe and acceptable under stated storage conditions.

It monitors sensory, chemical and microbiological quality changes over time.

Results guide date marking, packaging choice and storage instructions.

11. Explain the main steps involved in process design.

Define product specifications and select suitable ingredients, equipment and processing operations.

Establish process parameters such as time, temperature and mixing conditions.

Conduct trials, validate quality and safety, and standardise the process for scale-up.

12. Explain the contribution of novel food ingredients and processing technologies to product development.

Novel ingredients can improve nutrition, functionality, texture or flavour, or meet specific consumer needs.

Advanced processing technologies may improve safety, quality, convenience or shelf life.

Assess ingredients and technologies for safety, regulatory compliance, cost and consumer acceptance.

13. Explain the relationship between product design and formulation in new food product development.

Product design defines the concept, target consumer, sensory attributes, format and positioning.

Formulation specifies the ingredients and proportions needed to achieve the desired characteristics.

Trials align the formulation with product specifications, safety, cost and consumer expectations.

14. Discuss the major technological advances that support new food product development.

Advanced processing methods, including high-pressure and other non-thermal approaches, can help preserve quality while ensuring safety.

Ingredient technologies and encapsulation enable new textures, flavours and nutritional features.

Modern packaging, rapid testing and digital tools support quality control and shelf-life improvement.

Part C – Long Essay Questions (10 marks each)

15. Evaluate the role of consumer trials, market testing and sensory, chemical and microbiological assessment in deciding whether a newly developed food product is ready for commercialization.

Consumer trials and market testing assess liking, purchase intention, acceptable price, packaging and likely demand. Feedback identifies improvements and helps estimate commercial potential.

Sensory and chemical assessment: sensory tests evaluate appearance, aroma, taste and texture.

Chemical tests assess relevant composition and quality indicators such as pH, moisture or oxidation.

Microbiological assessment verifies safety and stability during the intended shelf life.

Commercialization should proceed only when safety and quality requirements are met, consumers accept the product, and market and financial potential are reasonable.

16. Analyse the complete new food product development process from idea generation to commercialization. Explain the purpose of each stage and show how decisions at one stage influence the next stage.

Idea generation and screening: Generate ideas from consumer needs, trends, research and competitors; screen out unsuitable, unsafe or commercially unrealistic ideas.

Concept development and feasibility: Define the target market and product benefits; assess technical feasibility, ingredient availability, cost, regulations and market potential. These findings determine whether development should continue.

Formulation, process design and prototype testing: Select ingredients and processing conditions, prepare prototypes, and evaluate sensory, chemical and microbiological quality. Results guide reformulation and process changes.

Consumer and market testing: Obtain feedback on acceptability, price, packaging and purchase intention. Use results to refine the product and marketing plan.

Scale-up and commercialization: Standardise the process, verify safety and quality, finalise packaging and labelling, arrange production and distribution, and launch. Post-launch feedback informs future improvements.

17. Discuss the changes in marketing strategies, sales and profit from the introduction stage to the decline stage of a new food product.

Introduction: Sales are usually low while awareness and distribution are built. Promotion and launch costs are high, so profit may be low or negative. Marketing focuses on informing consumers and encouraging trial.

Growth and maturity: Sales rise as acceptance and distribution increase. Profit improves, but competition intensifies. Marketing emphasises differentiation, brand loyalty, competitive pricing and product variants.

Decline: Sales and profits fall because of changing preferences, alternatives or market saturation. A company may reduce costs, reposition or modify the product, target a smaller segment, or discontinue it.

18. Explain the importance of integrating market analysis and sensory analysis in food product development.

Market analysis identifies target consumers, needs, trends, competitors, price expectations and potential demand.

Sensory analysis measures product attributes such as taste, aroma, appearance and texture through structured testing and consumer preference studies.

Integration ensures that the product is both acceptable to consumers and competitive. Market findings guide sensory targets, while sensory results help refine formulation and support positioning, pricing and launch decisions.